

$$\Rightarrow 9x = 63 \Rightarrow x = 7.$$

Present age of Paras = $(4 \times 7 + 10)$ years = 38 years.

34. Let the man's age be x years.

Then, son's age = $(45 - x)$ years.

$$(x - 5)(45 - x - 5) = 34$$

$$\Rightarrow (x - 5)(40 - x) = 34$$

$$\therefore 40x - x^2 - 200 + 5x = 34$$

$$\Rightarrow x^2 - 45x + 234 = 0$$

$$\Rightarrow x^2 - 39x - 6x + 234 = 0$$

$$\Rightarrow x(x - 39) - 6(x - 39) = 0$$

$$\Rightarrow (x - 39)(x - 6) = 0$$

$$\Rightarrow x = 39 \text{ or } x = 6.$$

$$\therefore \text{Man's age} = 39 \text{ years.}$$

35. Let the man's age be 7x years. Then, son's age = 3x years.

$$\therefore 7x \times 3x = 756$$

$$\Rightarrow 21x^2 = 756$$

$$\Rightarrow x^2 = 36 = 6^2$$

$$\Rightarrow x = 6.$$

The ratio of their ages after 6 years = $(7x + 6) : (3x + 6)$

$$= (7 \times 6 + 6) : (3 \times 6 + 6)$$

$$= 48 : 24 = 2 : 1.$$

36. Let x years ago the ratio of their ages be 3 : 5.

$$\text{Then, } \frac{40 - x}{60 - x} = \frac{3}{5} \Rightarrow 3(60 - x) = 5(40 - x)$$

$$\Rightarrow 180 - 3x = 200 - 5x$$

$$\Rightarrow 2x = 20$$

$$\Rightarrow x = 10.$$

$$\therefore 10 \text{ years ago, their ages were in the ratio } 3 : 5.$$

37. Let A's age be 5x years. Then, B's age = 3x years.

$$\frac{5x - 4}{3x + 4} = \frac{1}{1} \Rightarrow 5x - 4 = 3x + 4 \Rightarrow 2x = 8 \Rightarrow x = 4.$$

$$\therefore \frac{\text{A's age 4 years hence}}{\text{B's age 4 years ago}} = \frac{5x + 4}{3x - 4}$$

$$= \frac{5 \times 4 + 4}{3 \times 4 - 4} = \frac{24}{8} = \frac{3}{1} = 3 : 1.$$

38. Let the man's age be 4x years. Then, his wife's age = 3x years.

$$\text{Then, } \frac{4x + 4}{3x + 4} = \frac{9}{7} \Rightarrow 7(4x + 4) = 9(3x + 4)$$

$$\Rightarrow 28x + 28 = 27x + 36 \Rightarrow x = 8.$$

Man's age = (4×8) years

= 32 years, Wife's age

= (3×8) years = 24 years.

Let they be married y years ago. Then,

$$\frac{32 - y}{24 - y} = \frac{5}{3} \Rightarrow 3(32 - y) = 5(24 - y)$$

$$\Rightarrow 96 - 3y = 120 - 5y$$

$$\Rightarrow 2y = (120 - 96) = 24$$

$$\Rightarrow y = 12.$$

So, they were married 12 years ago.

39. Let Neelam's age be 5x years and Shiny's age be 6x years.

$$\left(\frac{1}{3} \times 5x\right) : \left(\frac{1}{2} \times 6x\right) = 5 : 9 \Rightarrow \frac{5x}{3 \times 3x} = \frac{5}{9}.$$

Thus, Shiny's age cannot be determined.

40. Let the son's age 18 years ago be x years. Then, man's age 18 years ago = $3x$ years.

$$(3x + 18) = 2(x + 18)$$

$$\Rightarrow 3x + 18 = 2x + 36$$

$$\Rightarrow x = 18.$$

Sum of their present ages = $(3x + 18 + x + 18)$ years

$$= (4x + 36) \text{ years}$$

$$= (4 \times 18 + 36) \text{ years} = 108 \text{ years.}$$

41. Let Ronit's present age be x years.

Then, the man's age = $(x + 3x)$ years = $4x$ years.

$$4x + 8 = \frac{5}{2}(x + 8) \Rightarrow 8x + 16 = 5x + 40 \Rightarrow 3x = 24 \Rightarrow x = 8.$$

$$\therefore \text{Required ratio} = \frac{(4x + 16)}{(x + 16)} = \frac{(4 \times 8 + 16)}{(8 + 16)} = \frac{48}{24} = 2 \text{ times.}$$

42. Let Sakshi's age 1 year ago be x years. Then, Promila's age 1 year ago = $4x$ years.

$\therefore$ Sakshi's age now = $(x + 1)$ years, Promila's age now = $(4x + 1)$ years.

$$(4x + 1) + 6 = (x + 1 + 6) + 9$$

$$\Rightarrow 4x + 7 = x + 16$$

$$\Rightarrow 3x = 9$$

$$\Rightarrow x = 3.$$

Ratio of Promila's age and Sakshi's age now

$$= \frac{(4x + 1)}{(x + 1)} = \frac{13}{4} = 13 : 4.$$

43. Let son's age 10 years ago be x years. Then, man's age 10 years ago = $3x$ years.

Son's present age = $(x + 10)$ years, Man's present age = $(3x + 10)$ years.

$$(3x + 10) + 10 = 2(x + 10 + 10)$$

$$\Rightarrow 3x + 20 = 2(x + 20)$$

$$\Rightarrow 3x + 20 = 2x + 40$$

$$\Rightarrow x = 20.$$

Ratio of present ages of man and the son

$$= \frac{3x + 10}{x + 10} = \frac{3 \times 20 + 10}{20 + 10} = \frac{70}{30} = 7 : 3.$$

44. 16 years ago, let $T = x$ years and $G = 8x$ years.

After 8 years from now, $T = (x + 16 + 8)$ years and G

$$= (8x + 16 + 8) \text{ years.}$$

$$\therefore 8x + 24 = 3(x + 24)$$

$$\Rightarrow 8x - 3x = 72 - 24$$

$$\Rightarrow 5x = 48.$$

$$8 \text{ years ago, } \frac{T}{G} = \frac{x + 8}{8x + 8} = \frac{\frac{48}{5} + 8}{8 \times \frac{48}{5} + 8} = \frac{48 + 40}{384 + 40} = \frac{88}{424} = \frac{11}{53}.$$

45. Let their ages be x years and $(x + 10)$ years.

Then $(x + 10 - 15) = 2(x - 15)$